Basic Electronic Circuits

Alexandra Andersson – 23N309

Labo – 23N310

Cyberlearn: 26_HES-SO-VS_IP3.12_Basic Electronic Circuits

Key: ELNB2026

Planning

Sem.	Date (Lundi)	Cours (Lundi)	Labo (Mercredi)	Cours (Vendredi)
1	**2026-09-14**	Thevenin, Superposition		Diode + Zener 1
2	**2026-09-21**	Diode + Zener 2		BJT intro
3	**2026-09-28**	BJT bias	Lab 1: Zener	BJT small signal model
4	**2026-10-05**	BJT ampli 1 EC		BJT ampli 2 CC
5	**2026-10-12**	BJT ampli 3 Cascade	Lab 2: BJT + ampli	EXAM (1 h)
6	**2026-10-19**	OPAMP 1		OPAMP 2
	2026-10-26	Travail autonome		
7	**2026-11-02**	OPAMP 3	Lab 3: OPAMP lin/nonlin	OPAMP 4
8	**2026-11-09**	OPAMP 5		OPAMP non-ideal 1
9	**2026-11-16**	OPAMP non-ideal 2	Lab 4: OPAMP triangle	EXAM (1 h)
10	**2026-11-23**	FILTER 1		FILTER 2
11	**2026-11-30**	FILTER 3	Lab 5: Second order TF	FILTER 4
12	**2026-12-07**	FILTER 5		MOSFET
13	**2026-12-14**	Current sources 1 (IT)	Lab 6: Filter LP 4th order (IT+PC)	Current sources 2 (IT)
	2026-12-21	Vacances de Noël		
	2026-12-28			
	2027-01-04			
14	**2027-01-11**	Comparator circuits 1 (IT)		Comparator circuits 2 (IT)
15	**2027-01-18**	Switched Capacitor 1 (IT)	Lab 7: Switched Capacitor 3 (IT)	Switched Capacitor 2 (IT)
	2027-01-25	Travail autonome		
	2027-02-01	EXAMS		

NOTES automne 2026
NOTEN Herbst 2026

Examens Prüfungen	Labo/Quiz Labo/Quiz	Examen de semestre Semesterprüfung
30%	20%	50%

Examens: 16 octobre, 20 novembre

Quizzes pas complete = pert des note

Quizzen nicht abgeschlossen = Notenverlust

Pert des points / Punktverlust

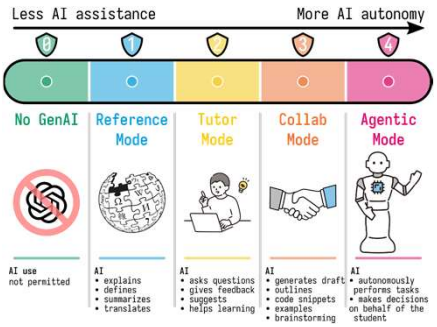
-0.01 par point inférieur à la réussite, à la note finale

-0.01 pro Punkt unter der Bestehensnote zur Endnote

Example / Beispiel					
	Quiz 1	**Quiz 2**	**Quiz 3**	**Quiz 4**	**Quiz 5**
Points to pass	9	9	10	8	11
Student scores	9 (90 %) ✓	10 (100 %) ✓	8 (66.67 %) ✗	…	5(45.45 %) ✗
	0	0	−0.02	−0.08	−0.06

Note après examen de semestre: 4.65, pertes: 0.16, note final: 4.49

Note nach Semesterprüfung: : 4.65, Verlust: 0.16, Schlussnote : 4.49



Cours	0
Quizzes Cyberlearn	0/1
Labo	0/1
Examens	0

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Vorlesungsinhalt
Table de matière du cours

Hes·so VALAIS WALLIS
Haute Ecole d'Ingénierie
Hochschule für Ingenieurwissenschaften

- Dioden, Gleichrichtung und StabilisierungBipolartransistor und Verstärker
- Operationsverstärker
- Nicht ideales Verhalten von Verstärkern
- Übertragungsfunktionen
- Aktive Filter
- MOSFETs
- Stromquellen und Komparator (IT)

- Diodes, rectification et stabilisation
- Transistor bipolaire et amplificateurs
- Amplificateurs opérationnels
- Comportement non-idéal d'amplificateurs
- Fonctions de transfert
- Filtres actives
- MOSFETs
- Sources de courant et comparateur (IT)

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Ch. 0: Rappel circuits RLC
Kp 0: Erinnerung RLC-Schaltungen

Loi d'Ohm / Ohmsches Gesetz

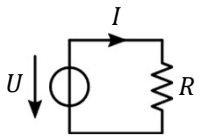
$U = I \times R$ ou/oder $V = I \times R$

Puissance/Leistung

$$P = I \times U$$

$$P = I^2 \times R$$

$$P = \frac{V^2}{R}$$

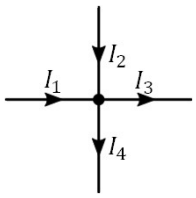


Lois de Kirchhoff
Kirchhoffsche Regeln

Loi des nœuds / Knotenpunktsatz

$$\sum I = 0$$

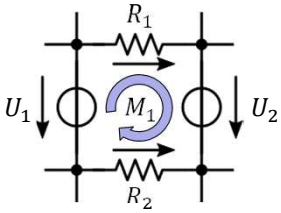
$$I_1 + I_2 - I_3 - I_4 = 0$$



Loi des mailles / Maschensatz

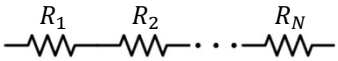
$$\sum U = 0$$

$$U_1 - U_{R1} - U_2 + U_{R2} = 0$$

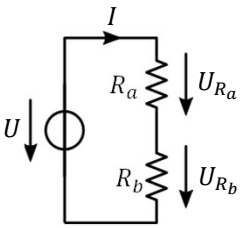


Résistances séries et diviseur de tension
Serienwiderstände und Spannungsteiler

$$R_{EQ} = R_1 + R_2 + \cdots + R_N$$



$$I = \frac{U}{R_a + R_b}$$



$$U_{R_a} = R_a I \qquad U_{R_b} = R_b I$$

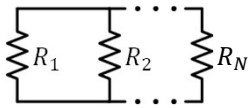
$$U_{R_a} = U\frac{R_a}{R_a + R_b} \qquad U_{R_b} = U\frac{R_b}{R_a + R_b}$$

$$U = U_{R_a} + U_{R_b}$$

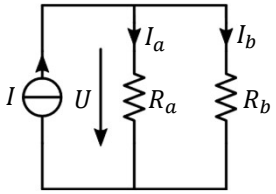
Résistances parallèles et diviseur de courant
Parallelwiderstände und Stromteiler

$$R_{EQ} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_N}}$$

$$R_{EQ} = \frac{R_1 R_2}{R_1 + R_2}$$



$$R_2 = \frac{R_1 R_{EQ}}{R_1 - R_{EQ}} \qquad R_1 = \frac{R_2 R_{EQ}}{R_2 - R_{EQ}}$$

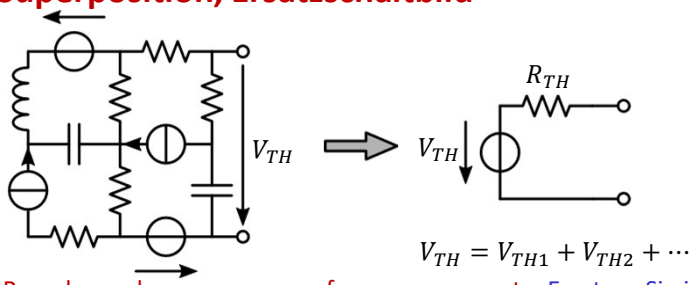


$$U = I\frac{R_a R_b}{R_a + R_b}$$

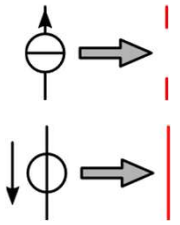
$$U = I_a R_a \qquad U = I_b R_b \qquad U = U_{R_a} = U_{R_b}$$

$$I_a = I\frac{R_b}{R_a + R_b} \qquad I_b = I\frac{R_a}{R_a + R_b} \qquad I = I_a + I_b$$

Superposition, circuit équivalent
Superposition, Ersatzschaltbild

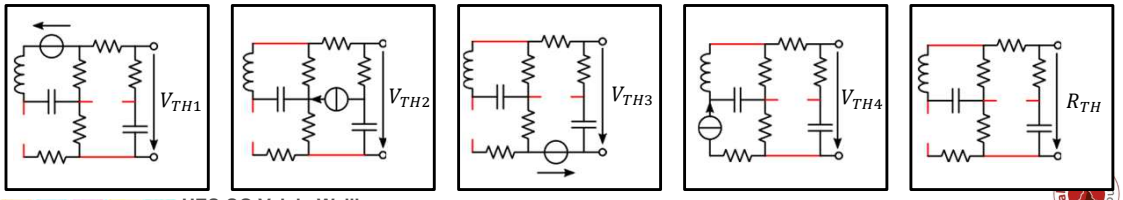


$$V_{TH} = V_{TH1} + V_{TH2} + \cdots$$



Remplacez chaque source sauf une par un court-circuit (source de tension) ou un circuit ouvert (source de courant)

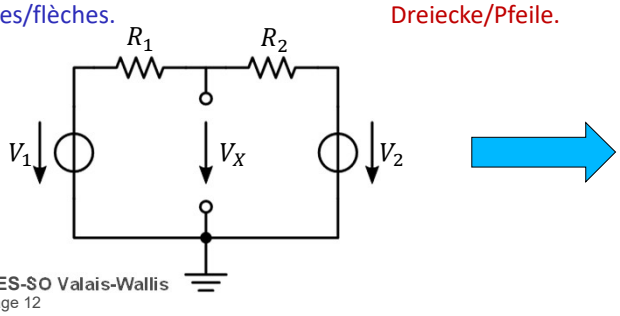
Ersetzen Sie jede Quelle bis auf eine mit einem Kurzschluss (Spannungsquelle) oder einem offenen Stromkreis (Stromquelle).



La masse / Die Masse

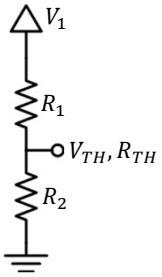
La tension existe uniquement en tant que *différence* de potentiel entre deux nœuds dans un circuit. Nous introduisons un nœud appelé *la masse*, et lui assignons une valeur de zéro. Quand nous parlons de la tension à un nœud, nous entendons sa *différence avec le nœud de masse*.
Souvent on remplace les sources des tensions qui constituent les alimentations avec des triangles/flèches.

Spannung existiert nur als eine *Potentialdifferenz* zwischen zwei Knoten in einer Schaltung. Wir stellen einen Knoten mit dem Namen *Masse* vor und weisen ihm den Wert Null zu. Wenn wir über die Spannung an einem Knoten sprechen, meinen wir seinen *Unterschied zum Masseknoten.*
Oft ersetzen wir die Spannungsquellen, aus denen die Netzteile bestehen, durch Dreiecke/Pfeile.



Diviseur de tension / Spannungsteiler

Exercice 0 / Aufgabe 0



$V_1 = 4\,V$
$R_1 = 33\,k\Omega$
$R_2 = 18\,k\Omega$

Calculez V_{TH} et R_{TH}.

Berechnen Sie V_{TH} und R_{TH}.

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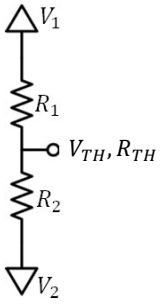
$V_1 = 4\,V$
$V_{TH} = 1.5\,V$
$R_{TH} = 20\,k\Omega$

Calculez R_1et R_2.

Berechnen Sie R_1 und R_2.

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Superposition: Exercice 1 / Aufgabe 1

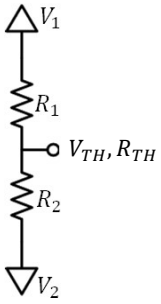


$V_1 = 5\ V$
$V_2 = -3\ V$
$R_1 = 1.2\ k\Omega$
$R_2 = 6.2\ k\Omega$

Calculez V_{TH} et R_{TH}.

Berechnen Sie V_{TH} und R_{TH}.

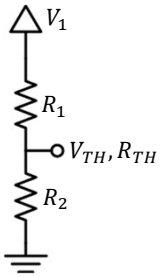
Superposition: Exercice 2 / Aufgabe 2



$V_1 = 1.2\ V$
$V_2 = 5\ V$
$V_{TH} = 3\ V$
$R_{TH} = 10\ k\Omega$

Calculez R_1 et R_2.

Berechnen Sie R_1 und R_2.



$$V_{TH} = \frac{R_2}{R_1 + R_2} V_1$$

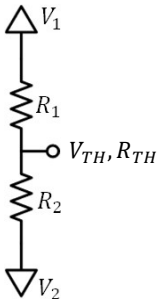
$$\boxed{V_{TH} = xV_1}$$

$$R_{TH} = \frac{R_1 R_2}{R_1 + R_2}$$

$$\boxed{x = \frac{R_2}{R_1 + R_2}}$$

$$\boxed{R_{TH} = xR_1}$$

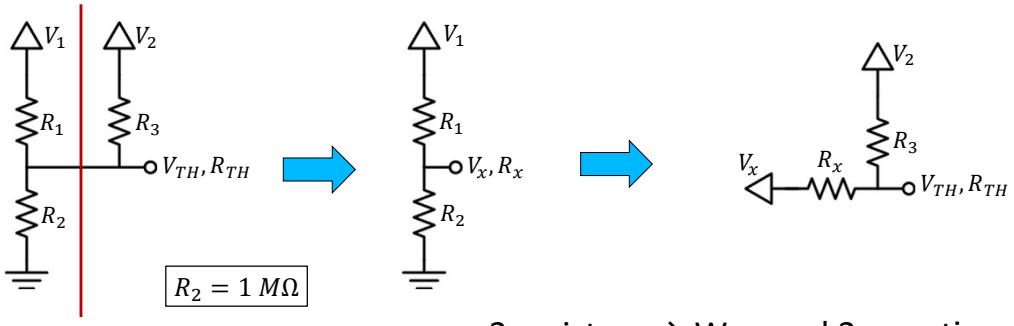
$$\boxed{R_2 = \frac{R_1 R_{TH}}{R_1 - R_{TH}}}$$



$$V_{TH} = \frac{R_2}{R_1 + R_2} V_1 + \frac{R_1}{R_1 + R_2} V_2$$

$$\boxed{V_{TH} = xV_1 + (1 - x)V_2}$$

Superposition: Exercice 3 / Aufgabe 3



3 resistors → We need 2 equations

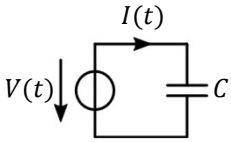
State A	State B
$V_1 = 1.5\ V$	$V_1 = 1.5\ V$
$V_2 = 0\ V$	$V_2 = 5\ V$
$V_{TH} = 0.5\ V$	$V_{TH} = 3\ V$

Calculez R_1 et R_3.

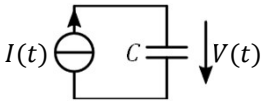
Berechnen Sie R_1 und R_3.

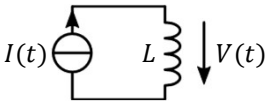
Condensateurs et Inductances dans le domaine temporel
Kondensatoren und Induktivitäten im Zeitbereich

dérivé et intégral / Derivat und Integral

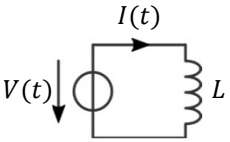


$$I(t) = C\frac{dV(t)}{dt} \qquad V(t) = \frac{1}{C}\int_{t_0}^{t} I(\tau)d\tau + V(t_0)$$

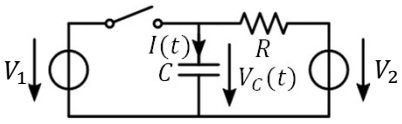




$$V(t) = L\frac{dI(t)}{dt} \qquad I(t) = \frac{1}{L}\int_{t_0}^{t} V(\tau)d\tau + I(t_0)$$



Condensateurs – approximation linéaire
Kondensatoren – lineare Näherung



$$V_C(0) = V_1$$

$$I(t) = \frac{(V_2 - V_C(t))}{R}$$

$$I(t) = C\frac{dV_C(t)}{dt}$$

$$RC\frac{dV_C(t)}{dt} = V_2 - V_C(t)$$

$$V_C(t) = A + B\exp(kt)$$

$$\frac{dV_C(t)}{dt} = kB\exp(kt)$$

$$RCkB\exp(kt) = V_2 - A + B\exp(kt)$$

$$RCkB\exp(kt) = B\exp(kt)$$

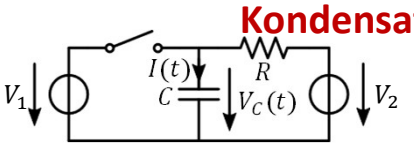
$$k = -\frac{1}{RC}$$

$$A = V_2$$

$$V_C(0) = A + B = V_1 \rightarrow B = V_1 - V_2$$

$$V_C(t) = V_2 + (V_1 - V_2)\exp\left(-\frac{t}{RC}\right)$$

Condensateurs – approximation linéaire
Kondensatoren – lineare Näherung



Série de Taylor / Taylorreihe

$$X(t) \approx X(t_0) + X'(t_0) \times (t - t_0) + \cdots$$

$$V_C(t) = V_2 + (V_1 - V_2)\exp\left(-\frac{t}{RC}\right)$$

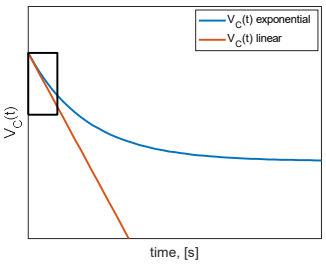
$$V_C(0) = V_1$$

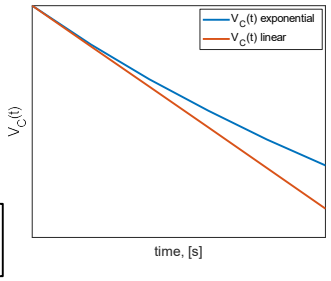
$$\frac{d}{dt}V_C(t) = -\frac{(V_1 - V_2)}{RC}\exp\left(-\frac{t}{RC}\right)$$

$$\frac{d}{dt}V_C(0) = -\frac{(V_1 - V_2)}{RC}$$

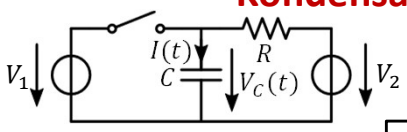
$$V_C(t) \approx V_1 - \frac{(V_1 - V_2)}{RC}\,t$$

$$\Delta V_C(t) \approx \frac{(V_1 - V_2)\Delta t}{RC}$$





Condensateurs – approximation linéaire
Kondensatoren – lineare Näherung



$$V_C(0) = V_1$$

$$I(0) = \frac{(V_2 - V_1)}{R}$$

$$I(t) \approx I(0) \quad t \text{ petit / klein}$$

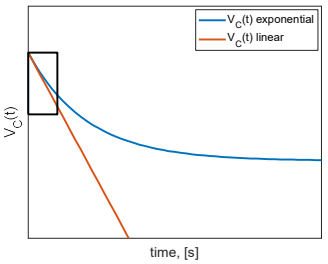
$$V_C(t) = \frac{1}{C}\int_{t_0}^{t} I(\tau)d\tau + V(t_0)$$

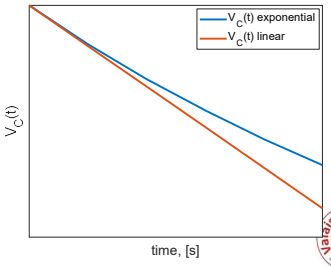
$$V_C(t) \approx \frac{1}{C}\,I(0)t + V_1$$

$$\Delta V_C \approx \frac{I(0)\Delta t}{C}$$

$$\Delta V_C \approx \frac{I_0\Delta t}{C} = \frac{(V_1 - V_2)\Delta t}{RC}$$
$$\Delta V_C \ll (V_1 - V_2)$$

$\frac{\Delta V_C}{V_1 - V_2}$ **lin**	$\frac{\Delta V_C}{V_1 - V_2}$ **exp**
1%	0.995%
2%	1.98%
5%	4.88%
10%	9.5%
20%	18%
25%	22%



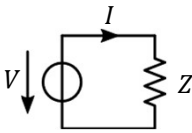


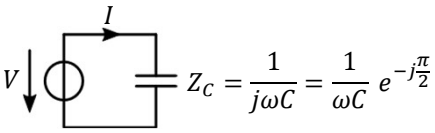
Circuits linéaires complexe – domaine fréquentiel
Komplexe lineare Schaltungen – Frequenzbereich

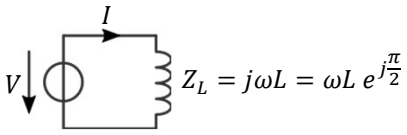
Loi d'Ohm / Ohmsches Gesetz

$$\tilde{V}(j\omega) = \tilde{I}(j\omega) \times Z\ (j\omega)$$

$$V(t) = \hat{V}\cos(\omega t + \phi) \rightarrow \tilde{V} = \hat{V}\ e^{j\phi}$$



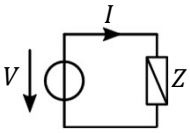




Puissance/Leistung

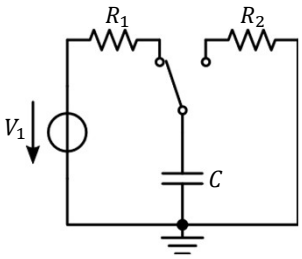
$$P = \tilde{V} \times \tilde{I}^*$$

konjugiert
conjugué



Général / Allgemein

Capacitance: Exercice / Aufgabe



$V_1 = 5\ V$
$V_C(t=0) = 2\ V$
$R_1 = 1\ k\Omega$
$R_2 = 10\ k\Omega$
$C = 10\ \mu F$
$T_1 = 1\ ms$
$T_2 = 5\ ms$

Schaltung:
1. Links : $T_1 = 1\ ms$
2. Rechts : $T_2 = 4\ ms$
3. Links : $T_1 = 1\ ms$
4. Rechts : $T_2 = 4\ ms$

Interrupteur
1. Gauche: $T_1 = 1\ ms$
2. Droite: $T_2 = 4\ ms$
3. Gauche: $T_1 = 1\ ms$
4. Droite: $T_2 = 4\ ms$

Condensateurs – approximation linéaire
Kondensatoren – lineare Näherung

Calculez V_C à $t = 1\ ms, t = 5ms, t = 6ms, t = 10ms$

Berechnen Sie V_C bei $t = 10\ ms, t = 5ms, t = 6ms, t = 10ms$